Probability & Statistics

APM_2F010_EP

Lecture Notes

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These are personal lecture notes and may contain errors. For the official course material consult the lecturer.

Contents

1 Lecture 1 **2**

2 Lecture 2 **3**
2.1 General Probability Spaces . . . 3
2.2 The Borel σ-algebra . . . 4
2.3 Random Variables on General Probability Spaces . . . 4
2.4 The Law (Distribution) of a Random Variable . . . 5
2.5 Conditional Probability and Independence . . . 5

3 Lecture 3 **7**
3.1 Important Distributions . . . 7
3.2 Relations Between Distributions . . . 8
3.3 Expectation . . . 9
3.4 Variance and Covariance . . . 11
3.5 L^p Spaces . . . 12

4 Lecture 4 **13**
4.1 Inequalities . . . 13
4.2 Moment Generating Function . . . 15
4.3 Cumulative distribution function . . . 16

5 Lecture 5 **18**
5.1 Applications of discrete random variables . . . 18

1 Lecture 1

Theorem 1: Fubini positive

Let $f : X \times Y \to [0, \infty)$, $g : X \to [0, \infty)$ and $h : Y \to [0, \infty)$ be piecewise continuous. Then:

$$\int_Y \left(\int_X f(x,y)g(x)dx \right) h(y)dy = \int_X \left(\int_Y f(x,y)h(y)dy \right) g(x)dx$$

- Works without integrability assumptions.
- Analogue: reordering double series with nonnegative terms.

Theorem 2: Fubini - Tonelli

Let $f : X \times Y \to \mathbb{R}$ (not necessarily positive), and let $g : X \to [0, \infty)$ and $h : Y \to [0, \infty)$ be piecewise continuous. If:

$$\int_Y \left(\int_X |f(x,y)|g(x)dx \right) h(y)dy < \infty$$

Then:

$$\int_Y \left(\int_X f(x,y)g(x)dx \right) h(y)dy = \int_X \left(\int_Y f(x,y)h(y)dy \right) g(x)dx$$

- Key assumption: integrable.

What is a probability?

Example 3: Rolling a fair dice

- Possible outcomes $\Omega = \{1, \dots, 6\}$.
- All outcomes have probability $1/6$.
- An event is "The result is odd".

Proposition 4: Discrete probability space

$(\Omega, \mathbb{P})$ is a discrete prob. space. $\mathbb{P}$ is called the discrete prob. on Ω. If $\forall A \subset \Omega$, $\mathbb{P}(A) = \frac{\text{Card}(A)}{\text{Card}(\Omega)}$, then $\mathbb{P}$ is called the uniform discrete probability.

Example 5: Non-discrete probability space

Let $E =$ "the extraction of a number in $[0, 1]$ uniformly at random". E is modeled with the uniform probability $\mathbb{P}$ on the space $\Omega = [0, 1]$.

$$\forall 0 \leq a \leq b \leq 1, \quad \mathbb{P}([a, b]) = b - a$$

$(\Omega, \mathbb{P})$ cannot be a discrete prob. space.

Definition 6: Probability function

We define a prob. function $\mathbb{P} : \mathcal{P}([0,1]) \to [0,1]$ s.t.

1. $\mathbb{P}([0,1]) = 1$
2. $\mathbb{P}$ is σ-additive
3. $\mathbb{P}([a,b]) = b - a \quad \forall a, b$

2 Lecture 2

2.1 General Probability Spaces

Definition 1: Sigma-algebra

Let Ω be an arbitrary set. A σ-algebra $\mathscr{A}$ is a subset of $\mathcal{P}(\Omega)$ such that:

1. $\emptyset \in \mathscr{A}$
2. If $A \in \mathscr{A}$, then $A^c \in \mathscr{A}$
3. If $(A_n)_{n\in\mathbb{N}}$ is a sequence of events in $\mathscr{A}$, then $\bigcup_{n\in\mathbb{N}} A_n$ is also in $\mathscr{A}$.

σ-algebras are subsets of $\mathcal{P}(\Omega)$ that are closed under the basic operations on sets.

Example 2: Sigma-algebras

Let Ω be an arbitrary set and $A \subseteq \Omega$. The following are σ-algebras:

- $\mathcal{P}(\Omega)$
- $\{\emptyset, A, A^c, \Omega\}$

Definition 3: General probability space

Let Ω be an arbitrary set, $\mathscr{A} \subseteq \mathcal{P}(\Omega)$ a sigma algebra and $\mathbb{P} : \mathscr{A} \to [0,1]$. The triplet $(\Omega, \mathscr{A}, \mathbb{P})$ is called a (general) probability space if:

1. $\mathbb{P}[\Omega] = 1$
2. If $(A_n)_{n\in\mathbb{N}}$ is a sequence of disjoint events in $\mathscr{A}$, then $\mathbb{P}\left[\bigcup_{n\in\mathbb{N}} A_n\right] = \sum_{n\in\mathbb{N}} \mathbb{P}[A_n]$.

If conditions (i) and (ii) are verified, $\mathbb{P}$ is called a (general) probability.

Example 4: General probability space

If $(\Omega, \mathbb{P})$ is a discrete probability space, then $(\Omega, \mathcal{P}(\Omega), \mathbb{P})$ is a general probability space.

Proposition 5: Basic properties of probability spaces

Let $\mathbb{P} : \mathcal{P}(\Omega) \to [0,1]$ be a probability and $A, B \subseteq \Omega$. Then:

1. Let $A_1, \ldots, A_n$ be a finite collection of pairwise disjoint events. Then $\mathbb{P}[A_1 \cup \cdots \cup A_n] = \sum_{j=1}^{n} \mathbb{P}[A_j]$. This property is called finite σ-additivity or additivity.
2. If $A \subseteq B$, $\mathbb{P}[B \setminus A] = \mathbb{P}[B] - \mathbb{P}[A]$. In particular, $\mathbb{P}[A] \leq \mathbb{P}[B]$ and $\mathbb{P}[A^c] = 1 - \mathbb{P}[A]$.
3. $\mathbb{P}[A \cup B] = \mathbb{P}[A] + \mathbb{P}[B] - \mathbb{P}[A \cap B]$. In particular, $\mathbb{P}[A \cup B] \leq \mathbb{P}[A] + \mathbb{P}[B]$.

2.2 The Borel σ-algebra

We need a σ-algebra which contains all intervals and is smaller than $\mathcal{P}(\Omega)$. The easiest choice is to take the smallest possible sigma algebra.

Definition 6: The Borel σ-algebra

Let $I \subseteq \mathbb{R}$ be an interval. The Borel sigma algebra $\mathcal{B}(I)$ is defined as:

$$\mathcal{B}(I) = \bigcap_{\{[a,b]:a,b\in I\}\subseteq\mathscr{A}\subseteq\mathcal{P}(I)} \mathscr{A}$$

To check that it is a good definition one has to verify that $\mathcal{B}(I)$ is indeed a sigma algebra.

Theorem 7: Uniform Probability Measure

There exists a general probability space $([0,1], \mathcal{B}([0,1]), \mathbb{P})$ such that $\mathbb{P}[[a,b]] = b - a$ for all $a, b \in \mathbb{R}$.

2.3 Random Variables on General Probability Spaces

Definition 8: Real random variable

Let $(\Omega, \mathscr{A}, \mathbb{P})$ be a probability space. A function $X : \Omega \to \mathbb{R}$ is called a real random variable if and only if for any $a, b \in \mathbb{R}$ we have that:

$$X^{-1}([a,b]) := \{\omega : X(\omega) \in [a,b]\} \in \mathscr{A}$$

A real random variable X defined on a general probability space $(\Omega, \mathscr{A}, \mathbb{P})$ is discrete if $X(\Omega)$ is at most countable.

Proposition 9: Discrete random variables as general r.v.

Let $(\Omega, \mathbb{P})$ be a discrete p.s. and $X : \Omega \to \mathbb{R}$ a r.v. Then X is a real random variable on $(\Omega, \mathcal{P}(\Omega), \mathbb{P})$.

Remark 10: Conventions

- If $E = \mathbb{R}$, we call X a real random variable; if $E = \mathbb{R}^n$ we call X a random vector.
- "Let X be a random variable" has to be read as "Let $(\Omega, \mathbb{P})$ be a discrete probability space and X be a random variable defined on it".
- We may denote events without brackets, and not specify the probability space. For instance: $\mathbb{P}[\{\omega : X(\omega) \in A\}] \Leftrightarrow \mathbb{P}[X \in A]$ and $\mathbb{P}[\omega] \Leftrightarrow \mathbb{P}[\{\omega\}]$.
- We may replace the intersection with a comma. For example: $\mathbb{P}[\{X \in A\} \cap \{Y \in B\}] \Leftrightarrow \mathbb{P}[X \in A, Y \in B]$.

2.4 The Law (Distribution) of a Random Variable

Proposition 11: The law of a random variable

Let X be a real valued r.v. The law (distribution) μ_X of X is the application $\mu_X : \mathcal{B}(\mathbb{R}) \to [0, 1]$ defined by $\mu_X(C) := \mathbb{P}[X^{-1}(C)]$. In this setting, the law $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_X)$ is a general probability space.

Remark 12: Two Kinds of Random Variables

In this course, a random variable $X : \Omega \to \mathbb{R}$ will be of one of two types:

- **Discrete random variables:** it exists a countable set $E \subset \mathbb{R}$ and $p_X : E \to \mathbb{R}_+$ s.t. $\forall C \in \mathcal{B}(\mathbb{R})$, $\mu_X(C) = \sum_{x \in C \cap E} p_X(x)$.
- **Absolutely continuous random variables:** it exists $f_X : \mathbb{R} \to \mathbb{R}_+$ s.t. $\forall a < b \in \mathbb{R}$, $\mu_X([a, b]) = \int_a^b f_X(u)du$.

Fundamental intuition: $\mathbb{P}(X \in [a - \varepsilon, a + \varepsilon]) = \mu_X([a - \varepsilon, a + \varepsilon]) \approx 2\varepsilon f_X(a)$. Here, p_X and f_X are respectively called the discrete density and the density of the random variable X.

2.5 Conditional Probability and Independence

Definition 13: Conditional probability

Let $(\Omega, \mathbb{P})$ be a discrete probability space and A, B two events such that $\mathbb{P}[B] > 0$. The conditional probability of A given B is denoted $\mathbb{P}[A|B]$ and defined by:

$$\mathbb{P}[A|B] := \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]}$$

Definition 14: Independence of two events

Let $(\Omega, \mathbb{P})$ be a discrete probability space and A, B two events. We say that A, B are independent if $\mathbb{P}[A \cap B] = \mathbb{P}[A]\mathbb{P}[B]$. If $\mathbb{P}[B] > 0$, we have the equivalent definition $\mathbb{P}[A|B] = \mathbb{P}[A]$.

Definition 15: Joint and marginal law

Let $X : \Omega \to E$, $Y : \Omega \to F$ be two r.v. The joint law $\mu_{X,Y}$ is the law of the random vector (X, Y). In particular, for all $C_1, C_2 \in \mathcal{B}(\mathbb{R})$:

$$\mu_{X,Y}(C_1 \times C_2) = \mathbb{P}[(X, Y) \in C_1 \times C_2] = \mathbb{P}[X \in C_1, Y \in C_2]$$

In this context μ_X, μ_Y are called marginal laws. We will write $p_{X,Y}$ and $f_{X,Y}$ for respectively the discrete joint density and the joint density according to the case we study.

Proposition 16: Marginal probabilities from joint density

Let $(\Omega, \mathbb{P})$ be a discrete probability space and $X : \Omega \to E$, $Y : \Omega \to F$ two random variables. Then:

$$\forall y \in F, \quad p_Y(y) = \sum_{x \in E} p_{X,Y}(x, y)$$
$$\forall x \in E, \quad p_X(x) = \sum_{y \in F} p_{X,Y}(x, y)$$

Definition 17: Independence of random variables

Let $(X_1, \ldots, X_N)$ be real valued random variables defined all on the same p.s. $(\Omega, \mathscr{A}, \mathbb{P})$. We say that they are independent if and only if for all possible choices of $[a_1, b_1], \ldots, [a_N, b_N]$ we have:

$$\mathbb{P}[X_1 \in [a_1, b_1], \ldots, X_N \in [a_N, b_N]] = \prod_{i=1}^{N} \mathbb{P}[X_i \in [a_i, b_i]]$$

An arbitrary family $(X_i)_{i \in I}$ of random variables defined on the same probability space is independent if and only if each finite subfamily is independent.

Proposition 18: Independence under complementation

Let $(A_i)_{i \in I}$ be independent events and $I' \subseteq I$. Then the events $(B_i)_{i \in I}$ defined by

$$B_i = \begin{cases} A_i, & i \in I' \\ A_i^c, & i \in I \setminus I' \end{cases}$$

are independent.

Proposition 19: Independence and joint density

Let $X : \Omega \to E_1, Y : \Omega \to E_2$ be two random variables defined on the same discrete probability space $(\Omega, \mathbb{P})$. Then X and Y are independent if and only if:

$$\forall x \in E_1, y \in E_2, \quad p_{X,Y}(x, y) = p_X(x) p_Y(y)$$

Proposition 20: Functions of independent random variables

Let X, Y be two independent random variables taking values in E and F respectively. Moreover, let H, K be two other sets and $f : E \to H$ and $g : F \to K$ two functions. Then the random variables $f(X)$ and $g(Y)$ are independent.

Proposition 21: The sum of independent random variables

Let X, Y be two absolutely continuous and independent random variables whose densities are f_X and f_Y. Then, the random variable $X + Y$ is absolutely continuous. Its density f_{X+Y} is given by the convolution $f_X * f_Y$ of f_X and f_Y, defined by:

$$(f_X * f_Y)(z) = \int_{-\infty}^{+\infty} f_X(x) f_Y(z - x) dx$$

It is also true in the discrete case:

$$p_{X+Y}(z) = \sum_{x \in \mathbb{R}} p_X(x) p_Y(z - x)$$

Be careful, the notation $\sum_{x \in \mathbb{R}}$ is only licit because this is the discrete case, this is highly wrong to write it in the absolutely continuous case.

3 Lecture 3

3.1 Important Distributions

Definition 1: Bernoulli Distribution

X is a Bernoulli random variable of parameter p if $p_X(1) = p$ and $p_X(0) = (1 - p)$. In this case, the distribution of X is the Bernoulli distribution of parameter p and we write $X \sim \text{Be}(p)$.

Definition 2: Geometric Distribution

X is a geometric random variable of parameter p if $p_X(0) = 0$ and $p_X(n) = p(1 - p)^{n-1}$ for $n \geq 1$. In this case, the distribution of X is the geometric distribution of parameter p and we write $X \sim \text{Geo}(p)$.

Definition 3: Binomial Distribution

X is a Binomial random variable of parameters N and p if

$$p_X(k) = \binom{N}{k} p^k (1 - p)^{N-k}, \quad \forall k = 0, 1, \ldots, N.$$

In this case, the distribution of X is the binomial distribution of parameters N and p and we write $X \sim \text{Bin}(N, p)$.

Definition 4: Poisson Distribution

X is a Poisson random variable of parameter λ, if

$$\forall n \in \mathbb{N}, \quad p_X(n) = \exp(-\lambda)\frac{\lambda^n}{n!}$$

In this case, the distribution of X is the Poisson distribution of parameter λ and we write $X \sim \text{Poi}(\lambda)$.

3.2 Relations Between Distributions

Proposition 5: From Bernoulli to Binomial

Let $(X_i)_{i\geq 1}$ be a sequence of i.i.d. random variables such that $X_1 \sim \text{Be}(p)$. Then $\sum_{i=1}^{N} X_i \sim \text{Bin}(N, p)$.

Proof. Let $X = \sum_{i=1}^{N} X_i$. We have:

$$\mathbb{P}(X = k) = \mathbb{P}\left(\sum_{i=1}^{N} X_i = k\right) = \mathbb{P}\left(\bigcup_{K\subset\{1,\dots,N\},|K|=k} \{X_i = 1 \forall i \in K, X_i = 0 \forall i \notin K\}\right)$$

By σ-additivity, since the sets are disjoint:

$$= \sum_{|K|=k} \prod_{i\in K} \mathbb{P}(X_i = 1) \prod_{i\notin K} \mathbb{P}(X_i = 0)$$

$$= \binom{N}{k} p^k (1-p)^{N-k}$$

□

Proposition 6: From Bernoulli to Geometrical

Let $(X_i)_{i\geq 1}$ be a sequence of i.i.d. random variables such that $X_1 \sim \text{Be}(p)$. Then $T = \inf\{i \geq 1 : X_i = 1\}$ is a geometric random variable of parameter p, i.e., $T \sim \text{Geo}(p)$.

Proof. First, we find the probability that $T \geq n$:

$$\mathbb{P}(T \geq n) = \mathbb{P}(X_i = 0 \; \forall i \in \{1, \dots, n-1\}) = (1-p)^{n-1}$$

Then, using the difference of probabilities:

$$\mathbb{P}(T = n) = \mathbb{P}(T \geq n) - \mathbb{P}(T \geq n+1) = (1-p)^{n-1} - (1-p)^n$$

$$= (1-p)^{n-1}(1-(1-p)) = p(1-p)^{n-1}$$

□

Proposition 7: Law of Small Numbers (simplified version)

Let $\lambda > 0$ and $(X_n)_{n \geq \lceil \lambda \rceil}$ be a sequence of random variables such that $X_n \sim \text{Bin}(n, \lambda/n)$. In particular, this means $\mathbb{E}[X_n] = \lambda$ for all n. Then we have

$$\lim_{n \to +\infty} p_{X_n}(k) = e^{-\lambda} \frac{\lambda^k}{k!} = p_X(k), \text{ with } X \sim \text{Poi}(\lambda).$$

Proof. Let $p = \lambda/n$. For a given r:

$$\mathbb{P}(X_n = r) = \binom{n}{r} \left(\frac{\lambda}{n}\right)^r \left(1 - \frac{\lambda}{n}\right)^{n-r}$$

$$= \frac{n(n-1)\dots(n-r+1)}{r!} \frac{\lambda^r}{n^r} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-r}$$

As $n \to \infty$, we analyze the terms:

- $\frac{n(n-1)\dots(n-r+1)}{n^r} \to 1$
- $\left(1 - \frac{\lambda}{n}\right)^n \to e^{-\lambda}$
- $\left(1 - \frac{\lambda}{n}\right)^{-r} \to 1$

Thus, $\mathbb{P}(X_n = r) \to e^{-\lambda} \frac{\lambda^r}{r!}$. □

3.3 Expectation

Definition 8: Expectation

Let X be a non negative valued random variable defined on a general probability space $(\Omega, \mathbb{P})$. We write $\mathbb{E}[X]$ the expectation of X which equals:

- $\sum_{x \geq 0} x p_X(x)$ in the discrete case,
- $\int_{\mathbb{R}} x f_X(x) dx$ in the absolute continuous case.

If X is a real valued random variable, we say that the expectation $\mathbb{E}[X]$ of X exists if one of the following conditions $\mathbb{E}[\mathbb{1}_{X \geq 0} X] < +\infty$ or $\mathbb{E}[\mathbb{1}_{X \leq 0}(-X)] < +\infty$ is verified. In that case the expectation value of X is given by

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{1}_{X \geq 0} X] - \mathbb{E}[\mathbb{1}_{X \leq 0}(-X)].$$

Example 9: Expectation Examples

- **Fair dice:** If $p_X(x) = 1/6$ for $x \in \{1, 2, 3, 4, 5, 6\}$, then $\mathbb{E}[X] = 3.5$.
- **Uniform law on** $[0, 1]$**:** If $f_X = \mathbb{1}_{[0,1]}$, then $\mathbb{E}[X] = 0.5$.

Proposition 10: Expectation - Equivalent Definition

Let $X : \Omega \to E$ be a discrete random variable defined over a discrete probability space $(\Omega, \mathbb{P})$. We have that $\mathbb{E}[X]$ exists if and only if one of the two conditions

$$\sum_{\omega\in\Omega, X(\omega)\geq 0} X(\omega)\mathbb{P}[\{\omega\}] < +\infty, \qquad \sum_{\omega\in\Omega, X(\omega)<0} X(\omega)\mathbb{P}[\{\omega\}] > -\infty$$

is verified. In this case

$$\mathbb{E}[X] = \sum_{\omega\in\Omega} X(\omega)\mathbb{P}[\{\omega\}]$$

Proof. Assume $X \geq 0$. Using the positive Fubini theorem:

$$\sum_{\omega\in\Omega} X(\omega)\mathbb{P}[\{\omega\}] = \sum_{x\in X(\Omega)} \sum_{\omega\in\Omega, X(\omega)=x} X(\omega)\mathbb{P}[\{\omega\}]$$

$$= \sum_{x\in X(\Omega)} x \sum_{\omega: X(\omega)=x} \mathbb{P}[\{\omega\}] = \sum_{x\in X(\Omega)} x\mathbb{P}(X = x) = \mathbb{E}[X]$$

The same logic extends to $\mathbb{E}[\mathbb{1}_{X\geq 0}X]$ and $\mathbb{E}[\mathbb{1}_{X\leq 0}(-X)]$. □

Proposition 11: Expectation of a Function of a Random Variable

Let $X : \Omega \to E$ be a random variable defined over a general probability space $(\Omega, \mathbb{P})$ and $g : E \to \mathbb{R}$ any function. In the discrete case, $\mathbb{E}[g(X)]$ exists if and only if either $\sum_{x:g(x)\geq 0} g(x)p_X(x) < +\infty$ or $\sum_{x:g(x)<0} g(x)p_X(x) > -\infty$ is verified. Then:

$$\mathbb{E}[g(X)] = \sum_{x\in X(\Omega)} g(x)p_X(x)$$

In the absolutely continuous case, if g is piecewise continuous, $\mathbb{E}[g(X)]$ exists if and only if either $\int_{\mathbb{R}} \mathbb{1}_{g(x)\geq 0} g(x)f_X(x)dx < +\infty$ or $\int_{\mathbb{R}} \mathbb{1}_{g(x)\leq 0} g(x)f_X(x)dx > -\infty$ is verified. Then:

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x)f_X(x)dx$$

Proof. For the discrete case, assuming $g \geq 0$:

$$\sum_{x\in X(\Omega)} g(x)p_X(x) = \sum_{q\geq 0} \sum_{x:g(x)=q} q\mathbb{P}(X = x)$$

$$= \sum_{q\geq 0} q\mathbb{P}(g(X) = q) = \mathbb{E}[g(X)]$$

□

Proposition 12: Product of Independent Random Variables

Let X, Y be two independent random variables. If $\mathbb{E}[X], \mathbb{E}[Y] < +\infty$, then $\mathbb{E}[XY] < +\infty$ as well and

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

PROOF. By Fubini positive, in the discrete case:

$$\mathbb{E}[XY] = \sum_x \sum_y xy\mathbb{P}(X = x, Y = y) = \sum_x \sum_y xy\mathbb{P}(X = x)\mathbb{P}(Y = y)$$

$$= \left(\sum_x x\mathbb{P}(X = x)\right)\left(\sum_y y\mathbb{P}(Y = y)\right) = \mathbb{E}[X]\mathbb{E}[Y]$$

□

Proposition 13: Properties of Expectation

Let X, Y two real valued random variables defined on the same general probability space $(\Omega, \mathbb{P})$. The following holds:

1. **Monotonicity:** If both $\mathbb{E}[X], \mathbb{E}[Y]$ exist, and $X(\omega) \leq Y(\omega)\ \forall \omega \in \Omega$, then $\mathbb{E}[X] \leq \mathbb{E}[Y]$.
2. **Absolute Value:** If $\mathbb{E}[X]$ exists, then $|\mathbb{E}[X]| \leq \mathbb{E}[|X|]$.
3. **Linearity:** If X, Y have finite expectation, and $a, b \in \mathbb{R}$, then $aX + bY$ has finite expectation and $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$.

PROOF. For discrete random variables:

1. $\mathbb{E}[X] = \sum_\omega X(\omega)\mathbb{P}[\{\omega\}] \leq \sum_\omega Y(\omega)\mathbb{P}[\{\omega\}] = \mathbb{E}[Y]$.
2. $|\mathbb{E}[X]| = |\sum_\omega X(\omega)\mathbb{P}[\{\omega\}]| \leq \sum_\omega |X(\omega)|\mathbb{P}[\{\omega\}] = \mathbb{E}[|X|]$ (by triangle inequality).
3. $\mathbb{E}[aX + bY] = \sum_\omega (aX(\omega) + bY(\omega))\mathbb{P}[\{\omega\}] = a\mathbb{E}[X] + b\mathbb{E}[Y]$.

□

Example 14: Uniform Random Variables Moments

If X is a uniform random variable over the interval $[a, b]$, then $f_X(x) = \frac{1}{b-a}$ for $x \in [a, b]$ and 0 otherwise.

$$\mathbb{E}[X] = \frac{1}{b-a}\int_a^b x dx = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}$$

$$\mathbb{E}[X^2] = \frac{1}{b-a}\int_a^b x^2 dx = \frac{b^3 - a^3}{3(b-a)} = \frac{b^2 + ab + a^2}{3}$$

3.4 Variance and Covariance

Definition 15: Variance

Let X be a real valued random variable with $\mathbb{E}[X^2] < +\infty$. The variance of X is

$$\mathbb{V}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Definition 16: Covariance

Let X, Y be two random variables s.t. $\mathbb{E}[|X|], \mathbb{E}[|Y|] < +\infty$ and $\mathbb{E}[|XY|] < +\infty$. The covariance of X and Y is

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

In particular, $\mathbb{V}(X) = \mathrm{Cov}(X, X)$.

Proposition 17: Properties of Variance and Covariance

Let X, Y be two random variables such that $\mathbb{E}[X^2], \mathbb{E}[Y^2] < +\infty$. Then we have

$$\mathbb{V}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2, \quad \mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

and

$$\mathbb{V}(X + Y) = \mathbb{V}(X) + \mathbb{V}(Y) + 2\mathrm{Cov}(X, Y).$$

- If $X \sim \mathrm{Be}(p)$, then $\mathbb{V}(X) = p - p^2$.
- If X is uniformly distributed on $\{-N, -N+1, \ldots, N-1, N\}$, then $\mathbb{V}(X) = \frac{N(N+1)}{3}$.

3.5 L^p Spaces

Definition 18: L^p spaces and norms

For any $p \geq 0$ we set

$$L^p(\Omega, \mathbb{P}) = \{X : \Omega \to \mathbb{R} : \mathbb{E}[|X|^p] < +\infty\}$$

Whenever there is no ambiguity, we abbreviate $L^p(\Omega, \mathbb{P})$ with L^p. If $X \in L^p$, its p-norm $\|X\|_p$ is defined as

$$\|X\|_p = (\mathbb{E}[|X|^p])^{1/p}$$

Theorem 19: Seminorm property

$\|\cdot\|_p$ is a seminorm on the space $L^p(\Omega, \mathbb{P})$. Recall that it verifies homogeneity ($\|\lambda x\| = |\lambda|\|x\|$) and the triangular inequality ($\|x + y\| \leq \|x\| + \|y\|$).

Proposition 20: Properties of L^p spaces

Let $(\Omega, \mathbb{P})$ be a general probability space and $1 \leq p \leq q < +\infty$. Then $L^p(\Omega, \mathbb{P})$ is a vector space:

- $X \in L^p \implies \lambda X \in L^p$
- $X, Y \in L^p \implies X + Y \in L^p$

If $(\Omega, \mathbb{P})$ is a discrete probability space:

1. If $0 < p \leq q < +\infty$, $L^p(\Omega, \mathbb{P}) \supseteq L^q(\Omega, \mathbb{P})$.
2. If $X, Y \in L^2(\Omega, \mathbb{P})$, then $XY \in L^1(\Omega, \mathbb{P})$.

Proof of L^p spaces properties.

- **Triangular inequality:** We can use the Minkowski inequality (the proof of it is not required for this course).
- **Proof of (i) $X \in L^q \implies X \in L^p$ for $p \leq q$:** If $X \in L^q$, then $\mathbb{E}[|X|^q] < +\infty$. Notice that if $|X| \leq 1$, then $|X|^p \leq 1$; otherwise, if $|X| > 1$, then $|X|^p \leq |X|^q$. Thus, in all cases, $|X|^p \leq 1 + |X|^q$. Taking the expectation yields $\mathbb{E}[|X|^p] \leq 1 + \mathbb{E}[|X|^q] < +\infty$, meaning $X \in L^p$.
- **Proof of (ii) $X, Y \in L^2 \implies XY \in L^1$:** Using the algebraic inequality $|xy| \leq \frac{x^2+y^2}{2}$, we get:

$$\mathbb{E}[|XY|] \leq \mathbb{E}\left(\frac{|X|^2 + |Y|^2}{2}\right) = \frac{\mathbb{E}[|X|^2] + \mathbb{E}[|Y|^2]}{2} < +\infty$$

□

4 Lecture 4

4.1 Inequalities

Proposition 1: Markov and Chebyshev's inequalities

Let X be a random variable and $\varepsilon > 0$.

- **Markov's inequality:** If X is non negative and in L^1:

$$\mathbb{P}[X \geq \varepsilon] \leq \frac{\mathbb{E}[X]}{\varepsilon}$$

- **Chebyshev's inequality:** If $X \in L^2$:

$$\mathbb{P}[|X - \mathbb{E}[X]| \geq \varepsilon] \leq \frac{\mathbb{V}(X)}{\varepsilon^2}$$

If $X \sim \text{Ge}(p)$, then Markov's inequality gives $\forall n \geq 0$, $\mathbb{P}[X \geq n] \leq \frac{1}{np}$.

Proof of Markov and Chebyshev's inequality. **Markov's inequality:**

$$\mathbb{P}(X \geq \varepsilon) = \mathbb{E}[\mathbb{1}_{X \geq \varepsilon}] \leq \mathbb{E}\left[\frac{X}{\varepsilon}\mathbb{1}_{X \geq \varepsilon}\right] \leq \frac{\mathbb{E}[X]}{\varepsilon}$$

Remark: You can apply any non-decreasing function $g(x)$. For example, $\mathbb{P}(X \geq \varepsilon) = \mathbb{P}(e^X \geq e^\varepsilon) \leq \frac{\mathbb{E}[e^X]}{e^\varepsilon}$.
Chebyshev's inequality: Using Markov's inequality on the random variable $(X - \mathbb{E}[X])^2$:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq \varepsilon) = \mathbb{P}(|X - \mathbb{E}[X]|^2 \geq \varepsilon^2) \leq \frac{\mathbb{V}(X)}{\varepsilon^2}$$

□

Example 2: Tails of a Poisson Po(λ)

Studying the tail of a random variable X means studying the function $t \mapsto \mathbb{P}[X \geq t]$ for large values of t. Recall that $\mathbb{E}[X] = \lambda$ and $\mathbb{V}(X) = \lambda$. By Markov's inequality:

$$\mathbb{P}[X \geq k] \leq \frac{\lambda}{k}$$

Thus, the tail decays at least as fast as k^{-1}. With Chebyshev's inequality we obtain for $k \geq \lambda$:

$$\mathbb{P}[X \geq k] \leq \mathbb{P}[|X - \mathbb{E}[X]| \geq k - \lambda] \leq \frac{\lambda}{(k-\lambda)^2}$$

This bound is better since it tells that the tail decays at least as fast as k^{-2}.

Proposition 3: Jensen's inequality

Let X be a random variable and φ a convex function. If both X and $\varphi(X)$ have finite expectation then

$$\mathbb{E}[\varphi(X)] \geq \varphi(\mathbb{E}[X])$$

Example 4: Norm inequalities

Recall: for $0 < p \leq q$, $X \in L^q \implies X \in L^p$. Let $q > 0$ and $X \in L^q$ and $p \leq q$. The function $x \mapsto x^{q/p}$ is convex. Therefore, using Jensen's inequality:

$$\mathbb{E}[|X|^q] \geq \mathbb{E}[|X|^p]^{\frac{q}{p}}$$

Raising both members to the power $1/q$ gives:

$$\mathbb{E}[|X|^q]^{\frac{1}{q}} \geq \mathbb{E}[|X|^p]^{\frac{1}{p}} \implies \|X\|_q \geq \|X\|_p$$

Proposition 5: Cauchy-Schwartz inequality

Let X, Y be two general random variables in L^2. Then $XY \in L^1$ and

$$\mathbb{E}[|XY|] \leq \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}$$

Equality holds ($\mathbb{E}[|XY|] = \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}$) if and only if there exists $t \in \mathbb{R}$ such that $X = tY$ almost surely, or $Y = tX$ almost surely.
Corollary: For all $X, Y \in L^2$, $\mathrm{Cov}(X, Y)$ is well defined and $|\mathrm{Cov}(X, Y)| \leq \sqrt{\mathbb{V}(X)\mathbb{V}(Y)}$.

Proof of Cauchy-Schwartz inequality. Consider the function $P(t) = \mathbb{E}((X + tY)^2)$. Expanding this gives:

$$P(t) = \mathbb{E}[X^2] + 2t\mathbb{E}[XY] + t^2\mathbb{E}[Y^2]$$

Since $(X + tY)^2 \geq 0$, we have $P(t) \geq 0$ for all $t \in \mathbb{R}$. Thus, $P(t)$ is a positive second-order polynomial in t, meaning its discriminant $\Delta \leq 0$.

$$\Delta = 4\mathbb{E}[XY]^2 - 4\mathbb{E}[X^2]\mathbb{E}[Y^2] \leq 0$$

$$\implies |\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2]}\sqrt{\mathbb{E}[Y^2]} = \|X\|_2\|Y\|_2$$

(If $\mathbb{E}[Y^2] = 0$, then $Y = 0$ almost surely, and the inequality is clearly true). □

4.2 Moment Generating Function

Definition 6: Moment generating function

Let X be a real valued random variable. The function $M_X : \mathbb{R} \to \mathbb{R}_+ \cup \{+\infty\}$ given by

$$M_X(t) = \mathbb{E}[\exp(tX)]$$

is called the moment generating function of X.

Example 7: Bernoulli random variables

If $X \sim \text{Be}(p)$, then $M_X(t) = (1-p) + p\exp(t)$, $\forall t \in \mathbb{R}$.

Theorem 8: Taylor series of MGF

Let X be a random variable and assume that for some $a > 0$ we have $M_X(t) < +\infty$ for all $t \in (-a, a)$. Then:

- The random variable X admits moments of all order (i.e., $\mathbb{E}[|X|^k] < +\infty$ for all $k \in \mathbb{N}$) and for all $t \in (-a, a)$, $M_X(t)$ is given by the Taylor series:

$$M_X(t) = \sum_{n=0}^{+\infty} \mathbb{E}[X^n] \frac{t^n}{n!}$$

- $M_X(t)$ is differentiable in $(-a, a)$ infinitely many times. Moreover, $M_X^{(n)}(0) = \mathbb{E}[X^n]$, where $M_X^{(n)}(0)$ is the n-th derivative of M_X evaluated at $t = 0$.

Proof of Theorem (Moment Generating Functions). Let $n \in \mathbb{N}_0$ and $t \in (0, a)$. We can bound the positive part by noting that $c_n = \sup_{x \geq 0} x^n e^{-tx} < +\infty$, which implies $\forall x \geq 0, x^n \leq c_n e^{tx}$. Therefore, $\mathbb{E}[|X|^n \mathbb{1}_{X \geq 0}] \leq c_n \mathbb{E}[e^{tX} \mathbb{1}_{X \geq 0}] \leq c_n M_X(t) < +\infty$. Similarly, for the negative part, letting $\tilde{c}_n = \sup_{x \leq 0} (-x)^n e^{tx}$, we get $\mathbb{E}[|X|^n \mathbb{1}_{X \leq 0}] = \mathbb{E}[(-X)^n \mathbb{1}_{X \leq 0}] \leq \tilde{c}_n M_X(-t) < +\infty$. Summing these, we conclude that $\mathbb{E}[|X|^n] = \mathbb{E}[|X|^n \mathbb{1}_{X \geq 0}] + \mathbb{E}[|X|^n \mathbb{1}_{X \leq 0}] < +\infty$, meaning X admits moments of all orders.
To obtain the Taylor series, we want to use the Fubini-Tonelli theorem to swap the expectation and the sum. This requires verifying that $\mathbb{E}\left[\sum_{n=0}^{\infty} \frac{|tX|^n}{n!}\right] < +\infty$, which is equivalent to $\mathbb{E}[e^{|tX|}] < +\infty$. We check this bound:

$$\mathbb{E}[e^{|tX|}] = \mathbb{E}[e^{tX} \mathbb{1}_{X \geq 0}] + \mathbb{E}[e^{-tX} \mathbb{1}_{X \leq 0}] \leq M_X(t) + M_X(-t)$$

Since $t \in (-a, a)$ implies $-t \in (-a, a)$, both $M_X(t)$ and $M_X(-t)$ are finite, meaning $\mathbb{E}[e^{|tX|}] < +\infty$. By Fubini-Tonelli, we can now safely interchange the expectation and the infinite sum:

$$M_X(t) = \mathbb{E}[e^{tX}] = \mathbb{E}\left[\sum_{n=0}^{\infty} \frac{(tX)^n}{n!}\right] = \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbb{E}[X^n]$$

Since the moment generating function $M_X(t) = \sum_{n=0}^{\infty} \frac{\mathbb{E}[X^n]}{n!} t^n$ converges for all $t \in (-a, a)$, it defines a power series with a radius of convergence $R \geq a$. By the standard properties of power series, any power series is infinitely differentiable strictly inside its interval of convergence, and its derivatives can be computed term-by-term. Taking the k-th derivative of $M_X(t)$ with

respect to t gives:

$$M_X^{(k)}(t) = \sum_{n=k}^{\infty} \frac{\mathbb{E}[X^n]}{(n-k)!} t^{n-k}$$

Evaluating this expression at $t = 0$, all terms in the sum where $n > k$ vanish because they are multiplied by 0^{n-k}. The only non-zero term is when $n = k$, which yields:

$$M_X^{(k)}(0) = \frac{\mathbb{E}[X^k]}{0!} 0^0 = \mathbb{E}[X^k]$$

This holds for all $k \in \mathbb{N}$, completing the proof. □

Example 9: MGF of Uniform and Geometric random variables

Uniform Distribution $\mathcal{U}(\{1, \ldots, N\})$: $p_X(k) = 1/N$ for $k \in \{1, \ldots, N\}$.

$$\mathbb{E}[e^{tX}] = \sum_{k=1}^{N} e^{tk} \frac{1}{N} = \frac{e^t}{N} \sum_{k=0}^{N-1} e^{tk}$$

For $t \neq 0$, using the geometric sum formula:

$$M_X(t) = \frac{e^t}{N} \frac{e^{tN} - 1}{e^t - 1}$$

As $t \to 0$, $M_X(t) \to 1$.
Geometric Distribution Ge(p): $p_X(k) = (1-p)^{k-1} p$ for $k \geq 1$.

$$\mathbb{E}[e^{tX}] = \sum_{k=1}^{\infty} e^{tk} p(1-p)^{k-1} = e^t p \sum_{k=0}^{\infty} (e^t(1-p))^k$$

This geometric series converges if and only if $e^t(1-p) < 1$, which is equivalent to $t < \log\left(\frac{1}{1-p}\right)$.

$$M_X(t) = \begin{cases} \frac{pe^t}{1-(1-p)e^t}, & \text{if } t < \log\left(\frac{1}{1-p}\right) \\ +\infty, & \text{if } t \geq \log\left(\frac{1}{1-p}\right) \end{cases}$$

To verify the limit as $t \to 0$ for the uniform distribution MGF:

$$M_X(t) = \frac{e^t}{N} \cdot \frac{e^{tN} - 1}{t} \cdot \frac{t}{e^t - 1}$$

As $t \to 0$: $\dfrac{e^{tN} - 1}{t} \to N$, $\dfrac{t}{e^t - 1} \to 1$ and $\dfrac{e^t}{N} \to \dfrac{1}{N}$. Therefore, the limit as $t \to 0$ is $\frac{1}{N} \cdot N \cdot 1 = 1$.

4.3 Cumulative distribution function

Definition 10: Cumulative distribution function

The cumulative distribution function of a random variable is defined by

$$\forall t \in \mathbb{R}, \quad F_X(t) = \mathbb{P}[X \leq t]$$

If X is a discrete r.v.:

$$F_X(t) = \sum_{x \leq t} p_X(x)$$

If X is an absolutely continuous r.v.

$$F_X(t) = \int_{-\infty}^{t} f_X(x)dx$$

In particular, it is continuous and of class $\mathcal{C}^1$ if f_X is continuous.

Proposition 11: Properties of CDF

The cumulative distribution function enjoys the following properties:

1. It is non decreasing
2. It is right-continuous $\lim_{y \downarrow x} F_X(y) = F_X(x)$
3. $\lim_{x \to +\infty} F_X(x) = 1$
4. $\lim_{x \to -\infty} F_X(x) = 0$

Continuity from above: Let $\mathbb{P} : \mathcal{P}(\Omega) \to [0,1]$ and let $(B_n)_{n \geq 1}$ be a non increasing sequence of events, i.e. $B_{n+1} \subseteq B_n \ \forall n \geq 1$. Then we have

$$\mathbb{P}\left[\bigcap_{n \geq 1} B_n\right] = \lim_{n \to +\infty} \mathbb{P}[B_n]$$

Proof of CDF Properties.

1. **Non decreasing:** If $t \leq t'$, then $\{X \leq t\} \subset \{X \leq t'\}$. Thus, $\mathbb{P}(X \leq t) \leq \mathbb{P}(X \leq t')$, which means $F_X(t) \leq F_X(t')$.
2. **Right-continuous:** Take a sequence (y_n) strictly decreasing to x. We set $B_n = \{X \leq y_n\}$. We have $B_n \supset B_{n+1}$. We claim that $\bigcap_{n \geq 0} B_n = \{X \leq x\}$. Let $\omega \in \bigcap_{n \geq 0} B_n$. Then $X(\omega) \leq y_n$ for all $n \geq 0$. Taking the limit gives $X(\omega) \leq x$, so $\omega \in \{X \leq x\}$. Conversely, if $X(\omega) \leq x$, since $x < y_n$, we have $X(\omega) \leq y_n$ for all $n \geq 0$, so $\omega \in \bigcap_{n \geq 0} B_n$. Applying continuity from above, $\mathbb{P}(X \leq y_n) \to \mathbb{P}(X \leq x)$, meaning $F_X(y_n) \to F_X(x)$. Because this holds for any such sequence (y_n), the function is right-continuous.
3. **Limit at $+\infty$:** Let $(x_n)_n$ be a sequence strictly increasing to $+\infty$. Let $B_n = \{X \leq x_n\}$. We have $\mathbb{P}(X \leq x_n) \to \mathbb{P}\left(\bigcup_{n \geq 0}\{X \leq x_n\}\right) = \mathbb{P}(X < +\infty) = 1$.
4. **Limit at $-\infty$:** Left as an exercise (similar proof using continuity from below).

□

Proposition 12: CDF for Discrete Random Variables

Let X be a discrete real valued random variable. Then

$$\forall x \in \mathbb{R}, \quad p_X(x) = F_X(x) - F_X(x^-)$$

where $F_X(x^-)$ denotes the left limit of F_X at x:

$$F_X(x^-) = \lim_{y \uparrow x} F_X(y)$$

Example 13: Bernoulli CDF

Let $X \sim \text{Be}(p)$. Then

$$F_X(x) = \begin{cases} 0, & \text{if } x < 0 \\ 1-p, & \text{if } 0 \leq x < 1 \\ 1, & \text{if } x \geq 1 \end{cases}$$

Proposition 14: Sufficient condition for absolute continuity

Let X be a real valued random variable. If F_X is piecewise $\mathcal{C}^1$, then X is an absolutely continuous random variable whose density is $f_X(x) = F_X'(x)$. *Piecewise $\mathcal{C}^1$ functions:* A function $g : (a, b) \to (-\infty, +\infty)$ is piecewise $\mathcal{C}^1$ if it is continuous on (a, b) and there exist $a = a_0 < a_1 < \cdots < a_N = b$ such that g is continuously differentiable on each interval (a_i, a_{i+1}), for all $i = 0, \ldots, N-1$.

Theorem 15: A general existence result

Let $F : \mathbb{R} \to [0, 1]$ be a non decreasing right-continuous function, such that $\lim_{x\to+\infty} F(x) = 1$ and $\lim_{x\to-\infty} F(x) = 0$. Then, there exists a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and a real valued random variable X defined on it such that $F_X = F$. Moreover, the density f_X of X is given by $f_X(x) = F_X'(x)$.

5 Lecture 5

5.1 Applications of discrete random variables

Problem 1: The coupon collector problem

We consider an album with N stickers and identify the album with the set $\{1, \ldots, N\}$. A collector can buy a sticker at the price of 1 Euro. Each sticker she buys is one of the possible N stickers with equal probability, independently from the previous purchases.
Let $(X_n)_{n\geq 1}$ be the sequence of purchased stickers, modeled as independent random variables all distributed according to the uniform law on $\{1, \ldots, N\}$. The random set $A_n := \{X_1, \ldots, X_n\}$ represents all the stickers collected in the first n purchases. We define $T_0 = 0$ and for $k = 1, 2, \ldots,$

$$T_k = \inf\{n \geq 1 : \text{Card}(A_n) = k\}$$

T_k is the time it takes to get k different stickers.

Solution to the Coupon Collector Problem. Let's analyze the time intervals between finding new stickers. $T_1 = 1$, since the first sticker is always new. The time to get the second new sticker is $T_2 - T_1 \sim \text{Geo}\left(\frac{N-1}{N}\right)$. In general, the time to get the $(k+1)$-th new sticker given you have k stickers is $T_{k+1} - T_k \sim \text{Geo}\left(\frac{N-k}{N}\right)$. These intervals $T_{k+1} - T_k$ are independent random variables.
We want to find information about T_N, the time until the collection is complete.

$$\mathbb{E}[T_N] = \mathbb{E}\left[\sum_{k=0}^{N-1} (T_{k+1} - T_k)\right]$$

By linearity of expectation:

$$\mathbb{E}[T_N] = \sum_{k=0}^{N-1} \mathbb{E}[T_{k+1} - T_k] = \sum_{k=0}^{N-1} \frac{N}{N-k} = N \sum_{p=1}^{N} \frac{1}{p}$$

As $N \to \infty$, the harmonic sum $\sum_{p=1}^{N} \frac{1}{p} \sim \log N$, so $\mathbb{E}[T_N] \sim N \log N$.
For the variance, since the intervals are independent, the variance of the sum is the sum of the variances:

$$\mathbb{V}(T_N) = \mathbb{V}\left(\sum_{k=0}^{N-1} (T_{k+1} - T_k)\right) = \sum_{k=0}^{N-1} \mathbb{V}(T_{k+1} - T_k)$$

□